ECE750T: Research Project Proposal

Reasoning Under Observation: Observer Effects on LLM Rationales

Yehia Fahmy | 20786266
University of Waterloo | Fall 2026

Q1: Research Question

How do observer identity and the declared visibility of a language model's verbalized reasoning affect answer accuracy, alignment with a misleading anonymous suggestion, and acknowledgment of that suggestion? I will compare a developer, a domain expert, and an automated AI monitor, each declared to see either the rationale and answer or only the answer. Oversight framing may change the evidence an audit collects, connecting evaluation and foundation-model safety.

Q2: Prior Work and Limitations

Turpin et al. [2023] and Chen et al. [2025] show that hints can influence answers without being acknowledged, but do not test this visibility comparison. Chen et al. [2026] study reasoning controllability, including explicit AI-monitoring awareness, rather than hint acknowledgment across observer roles. Covas and Hidalgo Toledo [2026] study framing in one debate topic; their human–AI observer contrast is suggestive rather than statistically resolved. Li et al. [2026] distinguish evaluation cues, recognition, and response; verbalized recognition is often not associated with behavioural change. These motivate a controlled extension, not a new claim of awareness.

Q3: Novelty of Approach

I extend hint-disclosure and monitoring-awareness studies by crossing observer identity with declared rationale visibility under a fixed misleading suggestion on a small local model. It tests framing effects when hints are present, not causal changes in hint susceptibility under every observer setting.

Q4: Impact and Motivation

If monitoring changes a rationale's acknowledgment of external suggestions, auditors may misjudge its transparency. Benchmark designers and safety evaluators could use the results to assess how oversight framing affects chain-of-thought audit evidence in this limited setting.

Q5: Risks and Assumptions

Hint alignment does not prove causal influence, and rationales do not establish internal computation or intentional concealment. Answer-only observer access is declared, not actual privacy or model belief. Matched templates limit wording differences but cannot remove role/authority confounding. Separate pilot items will check solvability, trace truncation, and labeling time; a separate comprehension probe checks the access instruction, not belief. Human validation will check automated labels; observer mentions may compromise blinding. Small samples, benchmark exposure, and one-model generalization limit conclusions.

Q6: Resources

I will download public BIG-Bench Hard data [Suzgun et al., 2023] and Qwen3-4B Q4_K_M from the official BBH and Qwen repositories. My available compute is an 8 GB RTX 3070 Ti; the model file is about 2.5 GB, but peak memory and throughput require a pilot measurement. The target is 120 main items total (40 per task), plus separate pilot items: 960 main generations. Python and `llama.cpp` will record native thinking traces and final labels with identical output fields, thinking mode, and decoding across arms. Context/output limits will be pilot-tested; no training or external compute is required.

Q7: Weekly Plan

At about five hours per week, I budget 34 scheduled hours plus roughly four hours of reserve through the provisional December 4 planning date. Within the first 8–10 project hours, I will check inference, scoring, trace access, and label reliability; if infeasible, I will reduce to one observer, two visibility states, and the two controls. Extra seeds and paraphrases are optional.

Week	**Plan**
Oct 12–18 (4h)	Refine literature; acquire data/model; build an AI-assisted inference/scoring smoke test.
Oct 19–25 (2h)	Check output formats and draft the acknowledgment rubric; keep midterm week light.
Oct 26–Nov 1 (5h)	Pilot tasks and labels; apply the feasibility check and freeze the protocol.
Nov 2–8 (4h)	Run the selected core design; check parsing, complete traces, and saved metadata.
Nov 9–15 (5h)	Validate automated acknowledgment labels against human coding; start paired analysis.
Nov 16–22 (5h)	Finish intervals and figures; complete a full report draft and freeze new experiments.
Nov 23–29 (4h)	Revise the report and prepare/rehearse slides around the micro-survey deadline.
Nov 30–Dec 4 (5h)	Finalize report, presentation, and reproducibility artifacts under final instructions.

Q8: Milestones and Success Criteria

The eight arms are three observers × rationale-visible/answer-only access, plus no-observer/no-hint and no-observer/hint controls. All generate the same rationale and answer fields in fresh conversations. Each hint is sampled uniformly from incorrect labels and fixed across hinted arms. The primary contrast is visibility, averaged over the three fixed roles; identity comparisons are exploratory. The primary endpoint is *unacknowledged wrong-hint alignment*: selecting the hinted label without explicitly mentioning the external suggestion, divided by all items in that arm. Alignment, acknowledgment, and accuracy will also be reported separately. No-hint answers will be scored against the same hidden target label to estimate excess alignment in the no-observer setting. Binary acknowledgment labels will be checked on 100–120 human-coded traces stratified across conditions and label predictions, using positive-class precision/recall; unreliable automation triggers the smaller, manually labeled design. Paired, task-stratified item bootstrap 95% intervals will quantify uncertainty under one seed schedule. Success is a completed reproducible study with checked scoring, validated labels, and documented limitations; null results may remain inconclusive about small effects.

Tools and Assistance

I used Cursor-supported language models for literature discovery, research planning, and proposal drafting. I will use my Cursor subscription and ChatGPT/Claude for implementation, debugging, and analysis assistance; automated labeler access will be arranged separately if needed. I will verify citations, code, labels, and results myself, document tool use in the final report, and retain human labels for validation.

References

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Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc Le, Ed Chi, Denny Zhou, and Jason Wei. Challenging BIG-bench tasks and whether chain-of-thought can solve them. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13003–13051. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.findings-acl.824. URL https://aclanthology.org/2023.findings-acl.824/.

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